DepthStream: A Free Lunch for Temporal Consistency in Video Depth

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Code and evaluation: `eval_sintel_v2.py` / `eval_bonn.py` / `bench_cpu.py`

Abstract

Monocular video depth is dominated by its per-frame formulation: every frame is inferred from scratch, so latency scales with model capacity and predictions flicker from frame to frame. The strongest remedy learns a temporal correction module around a large depth model — at the price of GPU-days of training and flagship-GPU inference. We show that this learning addresses a problem that distilled students do not have. Between consecutive frames the depth signal is nearly invariant; what changes is its transport and its affine parameterization. We therefore formulate streaming depth as the maintenance of an *anchored* depth field — transported between keyframes by backward optical flow, into which every fresh keyframe prediction is aligned by a closed-form scale–shift fit — and instantiate the formulation as DepthStream, a runtime without a single learned component, wrapping a frozen 6 M-parameter student (DepthART MobileNetV4-slim, TensorRT fp16). On Sintel (23 sequences) and Bonn RGB-D (9 sequences), streaming at keyframe interval $K{=}3$ matches per-frame accuracy exactly (δ_1 0.4614 vs. 0.4614; 99.96% retained on Bonn), raises the share of temporally consistent pixels from 55% to 73% (τ_5), and reduces latency by 20%. A decomposition of stream error into model and transport terms locates the amortization regime: accuracy holds to a cadence knee at $K^*{\approx}10$, is bounded by occlusion structure rather than flow accuracy, and is insensitive to temporal blending only under re-anchoring. A teacher cascade over the stream finds no correction headroom at 93.6–99.4% student–teacher agreement — the gap that learned propagation exploits on non-distilled bases is already closed by distillation. The runtime consequently runs in real time where per-frame inference cannot: a 2019 $99 Jetson Nano improves from 21 FPS to 90 FPS, a laptop CPU sustains 249 FPS, and accuracy is invariant across TensorRT-fp16 and ONNX-Runtime-fp32 backends ($|\Delta\delta_1| \leq 10^{-4}$).

1 Introduction

Video depth is dominated by its per-frame formulation: each frame is passed through a monocular model independently, so latency scales with model capacity and each output carries its own affine ambiguity and boundary noise. The costs are paid in full on every frame. Our reference teacher, Depth Anything V2-Large [2], needs 56 ms per frame on an RTX 3060 Laptop GPU at $1024{\times}436$; even the strongest propagation system, which learns to stabilize a foundation model, sustains 26 FPS on an RTX 3090 at half-megapixel resolution [3]; and independently predicted frames flicker, a jitter that every downstream consumer — segmentation, tracking, relighting, 3D capture — inherits.

Temporal correction is therefore bought with learning. VeloDepth [3] is the current state of this approach: a learned 55 M-parameter module corrects flow-warped depth and features around a frozen UniDepthV2 [21] base, trained for six days on $8{\times}$RTX 4090. The approach presumes that the discrepancy between consecutive predictions is complex enough to warrant a network.

For *distilled* students, we argue, it is not. Two properties of the distilled regime determine what a stream needs. First, *temporal invariance*: scene geometry is a slow variable, so the depth signal transports almost intact between consecutive frames; only its displacement and the model's own frame-to-frame noise differ. Second, *distillation closure*: a student distilled from a teacher agrees with that teacher on 93.6–99.4% of pixels, so the discrepancy a corrector could exploit is already closed. What remains between two frames is affine drift — a scale and shift — and transport error localized at occlusions. Neither calls for a network; both call for geometry.

We formalize streaming depth accordingly, as the maintenance of an *anchored* depth field: between keyframes the field is transported by backward optical flow [24]; at keyframes the fresh prediction of a frozen 6 M student (DepthART [1] MobileNetV4-slim) is aligned to the transported field by a closed-form scale–shift fit. The alignment re-parameterizes the new prediction into the stream's coordinate frame without overwriting it, which transfers temporal continuity while leaving the model's spatial error untouched. We instantiate this formulation as DepthStream (Fig. 2), a runtime with no learned component that exports unchanged to TensorRT and ONNX-Runtime.

The formulation makes claims that the paper tests individually. (i) On Sintel (23 sequences, 1,064 frames with ground-truth depth and flow [26]) and Bonn RGB-D (9 real sequences [27]), streaming at keyframe interval $K{=}3$ matches per-frame accuracy exactly (δ_1 0.4614 vs. 0.4614; 99.96% retained on Bonn) while raising the share of temporally consistent pixels from 55% to 73% and reducing latency by 20%: the per-frame baseline is strictly dominated. (ii) A decomposition of stream error into model and transport terms, evaluated by a cadence sweep over $K \in \{1, \ldots, 20\}$, locates the amortization regime: accuracy holds to a knee at $K^*{\approx}10$,

transport error is bounded by occlusion structure rather than flow accuracy, and temporal blending helps nowhere and hurts in occlusion-dense scenes. (iii) A teacher cascade over the stream finds no correction headroom, confirming the distillation argument quantitatively. (iv) The runtime runs in real time where per-frame inference cannot: on a 2019 $99 Jetson Nano, per-frame inference manages 21 FPS and the stream 90 FPS; on a laptop CPU the stream sustains 249 FPS; and accuracy is backend-invariant across TensorRT-fp16 GPU and ONNX-Runtime-fp32 CPU ($|\Delta\delta_1| \leq 10^{-4}$).

Contributions.

1. A formulation of streaming depth as anchored-field maintenance, with an error decomposition into model and transport terms that predicts when amortization dominates (Sec. 3).

2. DepthStream, a training-free runtime instantiating the formulation — keyframe inference, backward-flow transport, affine re-anchoring, residual-triggered scheduling — at 142–176 FPS on a 6 GB laptop GPU and 249 FPS on a CPU (Sec. 3, 4.6).

3. Strict-dominance and stability evidence on Sintel and Bonn RGB-D: streamed accuracy identical to per-frame inference at $K{=}3$, τ_5 $+33$–50%, and latency -20% (Sec. 4).

4. The finding that distillation closes the correction gap: a teacher cascade over distilled students has no headroom, which delimits when learned propagation [3] is rational — together with the cadence knee $K^* {\approx} 10$ and the occlusion bound on transport error (Sec. 4.7, 4.4).

Sec. 2 positions the work; Sec. 3 derives the runtime; Sec. 4 evaluates the claims; Sec. 5 and 6 discuss consequences and limitations.

2 Related Work

2.1 Single-image depth estimation: from relative to metric

Cross-dataset generalization, once the central obstacle of monocular depth, is now a solved enough problem that the open questions have moved elsewhere: to metric grounding and to camera awareness. MiDaS [10] established zero-shot *relative* depth by training on mixed datasets, and DPT [28] supplied the dense transformer decoder used by most successors; Depth Anything v1/v2 [11, 2] scaled the recipe with 62 M unlabeled images and became the de-facto open teacher — DAv2-L is the teacher in all our comparisons. Metric grounding arrived along two routes: ZoeDepth [12] combines a relative backbone with metric bins, Metric3D v2 [20] conditions a canonical-camera transform on intrinsics, and UniDepth [21] predicts camera rays and intrinsics jointly; diffusion models recast depth as generative translation (Marigold [13], DepthFM [14]) at seconds-per-frame cost. Two properties of this line matter here. All of it is *single-frame*: the temporal redundancy of video is invisible to its training objective, which is exactly the redundancy a stream can exploit. And its parameter count (10^8 scale) is inherited by whatever consumes it: these models define the accuracy ceiling that our teacher — and hence our student — operates under. DepthStream is deliberately agnostic to this line: any exportable per-frame model can be dropped into the runtime.

2.2 Lightweight depth models and edge deployment

The same distillation logic powers a separate line that compresses depth models for embedded inference. FastDepth [22] paired a mobile encoder with sparse-to-dense decoding for Jetson-class GPUs; RT-MonoDepth [15] redesigns encoder–decoder architectures for embedded latency; quantization and pruning toolchains push these further. DepthART [1] currently defines the *distilled* frontier for tiny models: it identifies two capacity-driven bottlenecks of naive compression (dataset-bias overfitting and unstable metric adaptation) and releases a deployment-ready family — including our student, the MobileNetV4-M-slim-SPF variant with pure standard operators, TensorRT and ONNX graphs. These works treat the model as the finished artifact and optimize a single quantity — the latency of one frame — so their cost scales linearly with frame rate and their outputs remain temporally uncalibrated. DepthStream is complementary: it adds the temporal dimension to any exportable per-frame model at negative latency cost, and our backend-invariance evaluation (Sec. 4) extends their deployment story to CPU-only devices.

2.3 Temporally consistent video depth

Every attempt to close the temporal gap pays somewhere: in inference-time compute, in optimization time, or in training time. *Post hoc smoothing* applies optical-flow consistency losses or filters to independently predicted frames [9] — cheap, but it attenuates the jitter without knowing which parts of the frame-to-frame difference are drift and which are real motion. *Per-video optimization* jointly solves depth, camera motion, and learned depth residuals over a whole clip [8], and practical variants reduce the cost for streaming capture [19]; these attain strong consistency but need the entire clip and minutes of optimization. *Learned video depth models* internalize temporal reasoning: NVDS stabilizes arbitrary per-frame models with a sequence-to-sequence network [4]; ChronoDepth [6] and DepthCrafter [5] use diffusion backbones for open-world long sequences; Video Depth Anything [7] distills a video-depth foundation for hour-long input. What none of these regimes separates is the affine component of the inconsistency — the part that costs nothing to fix at runtime. DepthStream requires no training at all: it is a runtime, not a model, and any of the above can serve as its student if exportable.

2.4 Propagation and amortized inference in dense prediction

Reusing per-frame computation across frames is a classic amortization strategy. Deep Feature Flow [16] propagated deep features from keyframes through flow for video recognition; XMem [17] used keyframe-anchored memory for long video segmentation. A related family adapts networks at test time to the incoming stream; our re-anchoring is orthogonal — it adapts only the two scalar degrees of freedom that the affine-invariant output leaves ambiguous, and never touches the weights. For depth, VeloDepth [3] is the closest system: a frozen UniDepthV2/UniK3D base runs on keyframes, and a *learned* 55 M-parameter propagation module (multi-branch ConvNeXt encoder, gated corrections, trained six days on 8×4090) refines flow-warped depth and features, reaching 26.2 FPS at 0.5 MP on an RTX 3090. Our runtime shares the keyframe and flow-warp skeleton but removes every learned component, targets *distilled tiny* students (6 M) instead of foundation models (409 M), and runs at 142–176 FPS on a 6 GB laptop GPU — an order of magnitude faster on a 7× smaller accelerator. The contrast also carries a conceptual finding (Sec. 4.7): VeloDepth's learned corrections exploit the disagreement between a non-distilled base and its judge; a student *distilled from that very judge* leaves almost no such space, which is why our zero-training runtime matches its consistency role. Our speculative-cascade analysis (Sec. 4.7) formalizes this via the test-time-selection analogy to speculative decoding in language models [18]: drafting by a cheap proposer pays only when a verifier would reject a measurable fraction of drafts — for distilled students, it would not.

3 Method

3.1 Problem formulation

Streaming depth asks for a sequence of depth fields $\{D_t\}$ given frames $\{I_t\}$ and a frozen per-frame student ϕ. Two error sources are distinguishable at inference time. The *model error* $\epsilon_m(t) = \epsilon(D_t^\phi, D_t^*)$ is the spatial error of the student against ground truth D_t^*; distillation fixes it, and it cannot be reduced without retraining. The *transport error* measures how well the previous field, once displaced, predicts the current model output up to an affine map:

$$\epsilon_w(t) \;=\; \min_{\alpha,\beta} \left\| \alpha\, \hat{D}_t + \beta - D_t^\phi \right\|, \qquad \hat{D}_t = \mathcal{W}(D_{t-1}, F_t), \tag{1}$$

where F_t is the inter-frame flow and $\mathcal{W}$ backward sampling. Per-frame inference pays ϵ_m on every frame. A stream pays ϵ_w on propagated frames and ϵ_m on keyframes, so its error — and, by the same interpolation, its latency (Sec. 3.7) — varies with the cadence K. Three questions define the design space: (i) how large may K grow before ϵ_w dominates ϵ_m; (ii) what bounds ϵ_w; (iii) whether the aligned model output should be blended with the propagated field. The runtime below answers these constructively; Sec. 4 measures them.

3.2 Setting and notation

The runtime is organized around the consequence of Eq. 1: re-anchoring must re-parameterize new observations into the anchor's frame without overwriting them, since the model error ϵ_m cannot be reduced at inference time. Let I_t denote the frame at time t of a monocular stream, and let $\phi : I \mapsto D^\phi$ be a frozen per-frame depth student (here, DepthART [1] with a MobileNetV4-M-slim backbone and a Single-Path-Fusion decoder, 6.05 M parameters), exported to TensorRT fp16. ϕ is metric-agnostic: its output is an affine-invariant relative depth. The runtime maintains the last *anchored* full-resolution depth field D_{t-1} and the index t_k of the last keyframe. At every frame exactly one of two paths produces D_t (Fig. 2).

3.3 Amortized scheduling

A frame is a *keyframe* if $t - t_k \geq K$ (cadence K) or if no anchored field exists yet; otherwise it is a *propagated* frame. Sec. 3.6 replaces the fixed cadence with an adaptive trigger. The keyframe path invokes ϕ once and upsamples its native 224^2 output to the working resolution; the propagation path reuses the anchored field.

3.4 Flow-based depth propagation

For a propagated frame we compute a backward optical flow $F_t : u \mapsto u + f_t(u)$ between I_t and I_{t-1} at a reduced resolution (256 px width), upsample it, and warp the previous field:

$$\hat{D}_t = \mathcal{W}(D_{t-1}, F_t), \tag{2}$$

where $\mathcal{W}$ is bilinear backward sampling. Two properties make this the right correspondence: the flow is *backward* (defined at the pixels of D_t, so no splatting or occlusion reasoning is needed for the sampling step), and it is applied to the *anchored* field, which already agrees in scale with all previous outputs. We use DIS (fast preset) [24]; ablations show that neither a higher processing resolution nor ground-truth forward flow improves accuracy (Sec. 4.5).

3.5 Affine re-anchoring

The student's output is affine-invariant, so the absolute scale and offset of D_t^ϕ are not comparable to the anchored field. We re-anchor by closed-form least squares over the valid pixels:

$$(\alpha_t, \beta_t) = \arg\min_{\alpha,\beta} \left\| \alpha\, \hat{D}_t + \beta - D_t^\phi \right\|_2^2, \qquad D_t = \alpha_t D_t^\phi + \beta_t. \tag{3}$$

Re-anchoring transfers the temporal continuity established by propagation to the fresh prediction — the field keeps the previous frame's scale — while discarding the propagated content itself. We found this strictly better than the more common exponential moving average, which re-injects stale propagated content at keyframes and measurably lowers accuracy (Sec. 4.5); the final runtime therefore uses $\lambda = 1$ equivalently, i.e., pure re-anchoring without blending.

Algorithm 1: DepthStream per-frame update (training-free)

Require: frame I_t; anchored field D_{t-1}; last key t_k

1: $g_t \leftarrow \mathrm{resize}(\mathrm{gray}(I_t))$
2: $due \leftarrow (t - t_k) \geq K$
3: **if** not due **and** D_{t-1} exists **then**
4: $\quad F_t \leftarrow \mathrm{Flow}(g_t, g_{t-1})$ ▷ backward, @256 px
5: $\quad \hat{D}_t \leftarrow \mathcal{W}(D_{t-1}, \mathrm{up}(F_t))$
6: $\quad$ **if** adaptive **and** $t - t_k \geq K_{\min}$ **then**
7: $\quad\quad \rho_t \leftarrow \mathrm{frac}(|\mathcal{W}(g_{t-1}, F_t) - g_t| > \theta)$
8: $\quad\quad due \leftarrow \rho_t > \eta$ ▷ flow-failure trigger
9: $\quad$ **end if**
10: **end if**
11: **if** due **then**
12: $\quad D_t^{\phi} \leftarrow \mathrm{up}_{224 \to H,W}(\phi(I_t))$
13: $\quad (\alpha, \beta) \leftarrow \mathrm{LSQ}(\hat{D}_t, D_t^{\phi})$ ▷ Eq. 3
14: $\quad D_t \leftarrow \alpha D_t^{\phi} + \beta; \quad t_k \leftarrow t$
15: **else**
16: $\quad D_t \leftarrow \hat{D}_t$
17: **end if**
18: **return** D_t

Figure 1: **DepthStream per-frame update.** Every frame takes exactly one of two paths: propagation through the anchored field (lines 3–10) or keyframe inference with affine re-anchoring (lines 11–14); the photometric-residual monitor (lines 6–9) promotes propagated frames to keyframes where flow fails. No line involves learning.

3.6 Photometric-residual adaptive scheduling

Figure 1 summarizes the complete per-frame decision procedure. Fixed-cadence keyframes cannot distinguish a static scene (propagation is reliable indefinitely) from a fast-panning one (flow fails within a frame). We monitor a photometric residual on the flow-resolution gray frames, $\rho_t = \frac{1}{|\Omega|}\big|\{u : |\mathcal{W}(g_{t-1}, F_t)(u) - g_t(u)| > \theta\}\big|$, with θ=0.15 in $[0, 1]$ intensity. If $\rho_t > \eta$ (η=0.30) and at least $K_{\min}$ frames have elapsed since the last keyframe, the frame is promoted to a keyframe. The monitor reuses the flow already computed for propagation and costs < 0.1 ms.

3.7 Complexity and deployment

Per propagated frame the runtime performs one small-resolution flow inference and one full-resolution warping pass; per keyframe it adds one student inference. Writing c_m for the model call, c_f for flow and c_w for warping, the expected per-frame cost under fixed cadence K is

$$c(K) \;=\; \tfrac{1}{K} c_m \;+\; \big(1 - \tfrac{1}{K}\big)(c_f + c_w), \qquad (4)$$

i.e., the model cost is amortized linearly while the propagation cost is paid on every frame. Table 7 measures the components: because c_m at 224^2 is dominated by Python-side preprocessing rather than GPU compute, amortization shifts the bottleneck to the flow term on every platform — most visibly on the CPU and Jetson, where the stream is *faster* than the GPU not despite but because of the tiny model. The entire runtime is expressible in standard operators (the frozen student has official ONNX graphs) and exports to TensorRT and ONNX-Runtime without custom plugins; on the Nano the same fp32 engine serves both the engine path and, through ONNX, any CPU-only fallback.

Re-anchoring itself costs one pass over the valid pixels ($<$ 1 ms at 1024×436 in NumPy) and its closed-form solution (Eq. 3) needs no iteration: with centered sums, $\alpha = \frac{\sum \Delta\hat{D}\,\Delta D^{\phi}}{\sum (\Delta\hat{D})^2}$ and $\beta = \overline{D^{\phi}} - \alpha\,\overline{\hat{D}}$, clipped to $\alpha \in [0.25, 4]$ to bound drift under degenerate overlaps (fewer than 100 valid pixels fall back to the identity, keeping the stream defined).

4 Experiments

4.1 Setup

Student and teacher. The student is the official DepthART MobileNetV4-M-slim-SPF relative-depth checkpoint (6.05 M parameters, 3.78 GMACs at 224^2) [1], exported to ONNX and built into a TensorRT fp16 engine (TRT 11.3). It is distilled by the authors of [1] from DAv2 outputs, which makes our teacher comparison genealogically strict: the teacher is Depth Anything V2-Large [2] (335 M, ViT-L), run per-frame in fp16 at a 518-px longest side.

Datasets. *Sintel* [26]: all 23 training sequences with depth annotations (1,064 frames, 1024×436, clean pass), providing ground-truth depth *and* forward flow. *Bonn RGB-D Dynamic* [27]: nine dynamic indoor sequences (balloon, balloon_tracking, crowd, kidnapping_box variants and others; 540 frames at 640×480) with Kinect ground truth — a real-sensor counterpoint to Sintel's synthetic content. RGB–depth pairs are matched by nearest timestamps; ground truth is capped at 500 m (Sintel sky outliers reach 10^{10} m) and 10 m (Kinect range), respectively.

Metrics. δ_1: fraction of pixels within a 1.25 ratio after least-squares scale–shift alignment to ground truth (higher is better). TAE: temporal warping error — consecutive predictions are aligned and compared after warping by ground-truth flow (Sintel) or estimated backward flow (Bonn; identical across all methods, so rankings are comparable) — normalized by the mean magnitude (lower is better). Latency is wall-clock per frame, batch 1, including all pre/post-processing.

Configs. *dense*: per-frame student. *kf3*, *kf10*: fixed keyframe cadence. *kf10-adp*: adaptive scheduling with maximum gap 10. *kf10-oracle*: propagation with ground-truth forward flow (available on Sintel only). *kf3-hiflow*: flow at 384-px width. *dav2l518*: the teacher.

4.2 Main results: streaming matches per-frame accuracy

Table 1 is the paper's core result; Fig. 3 shows the latency–accuracy plane.

Amortization is free at K=3. On Sintel, kf3 reproduces the per-frame student's δ_1 exactly (0.4614 vs. 0.4614) while reducing temporal error by 15.3% and latency by 20%; on

Table 1: **Main results.** Top: Sintel, 23 sequences / 1,064 frames (GT-flow TAE). Bottom: Bonn RGB-D, 3 dynamic sequences (estimated-flow TAE). All configs share the student and differ only in scheduling/propagation.

Sintel (synthetic, 23 seq / 1,064 frames)			
Config	$\delta_1\uparrow$	TAE$\downarrow$	ms/frame$\downarrow$
DAv2-L @518 per-frame	**0.4930**	0.1844	56.4
DepthART dense per-frame	0.4614	0.1585	10.6
DepthStream kf3	**0.4614**	0.1343	8.4
DepthStream kf10	0.4406	0.1094	6.1
DepthStream kf10-adp	0.4434	**0.1054**	7.6
Bonn RGB-D (real, 9 seq / 540 frames)			
Config	$\delta_1\uparrow$	TAE$\downarrow$	ms/frame$\downarrow$
DAv2-L @518 per-frame	**0.9924**	–	85.6
DepthART dense per-frame	0.9866	**0.4668**	12.2
DepthStream kf3	0.9862	0.4752	7.0
DepthStream kf10-adp	0.9811	0.4759	5.8

Bonn (9 sequences) it keeps 99.96% of the per-frame accuracy at 43% lower latency. There is no accuracy–consistency trade-off at moderate intervals — there is a strict improvement. The identical accuracy is itself informative: it is the empirical signature that transport error is affine-absorbed at this cadence, so streaming alters the temporal spectrum of the output without touching its spatial content — precisely the separation that Eq. 1 assumes.

Streaming rivals the teacher. At $1/6.7$ (Sintel) and $1/12.2$ (Bonn) of the teacher's latency, kf3 retains 93.6% and 99.4% of the teacher's δ_1, and its temporal error is $1.37\times$ *lower* than the teacher's per-frame error on Sintel (0.1343 vs. 0.1844). The pattern strengthens with adaptive scheduling: kf10-adp achieves the best temporal error of all configurations (0.1054, $1.75\times$ better than the teacher) at $1/7.4$ of its cost. That a stream of one model is more stable than a second, stronger model queried per frame is worth pausing on: temporal error is not a function of model quality alone, and per-frame inference does not optimize it at all.

Propagation decay is shallow. Fig. 4 plots δ_1 against distance from the last keyframe: kf10-adp decays only 3.3% over nine propagated frames (0.4695 to 0.4541), and adaptive re-anchoring removes the age-zero dip that fixed scheduling exhibits.

4.3 Temporal stability, revisited with τ_5

Table 2 lists per-sequence accuracy for all configurations; per-sequence variance (0.03 in ambush_2 to 0.72 in market_6) is dominated by scene content, and the streaming configurations track their dense counterpart within ± 0.02 on 21 of 23 sequences.

TAE measures the magnitude of frame-to-frame drift; the propagation literature additionally reports τ_5 [3], a per-pixel *inlier rate*: depth from $t-1$ is warped to t (GT flow), scale–shift aligned per pair, and a pixel counts as consistent when $|D_t - \tilde{D}_{t-1}| < 5\%$ of the depth. We implement τ_5 without

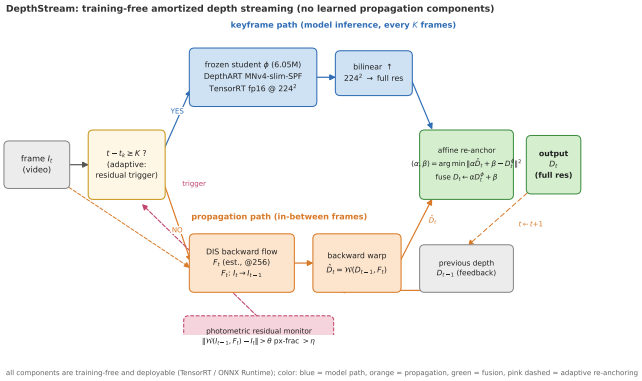


Figure 2: **DepthStream runtime.** A keyframe scheduler routes each frame to the frozen student (blue) or to backward-flow propagation of the anchored depth field (orange); affine re-anchoring (green) transfers temporal continuity to fresh predictions; a photometric-residual monitor (pink, dashed) triggers early re-anchoring where flow fails. No component is trained.

the ego-motion term (our runtime consumes no poses; the scale–shift alignment absorbs the dominant correction) and report it for all Sintel configurations in Table 4. The ordering matches TAE exactly, but the effect size is more readable and speaks directly to consumers of depth: under per-frame inference fewer than six pixels in ten agree with their transported predecessors — a video-depth stream that a tracking or fusion stage can rely on — whereas streaming raises that to seven in ten at $K{=}3$ and beyond eight in ten at $K{=}10$, a $1.5\times$ improvement that survives the switch to the rival metric family. Per-frame DAv2-L scores 0.5516 under the same protocol.

4.4 Cadence sweep

Fig. 5 sweeps the cadence $K \in \{1, \ldots, 20\}$ over all 23 Sintel sequences. Latency falls steeply and then asymptotes ($12.4 \rightarrow 5.1$ ms, a $2.4\times$ reduction bounded below by the flow term of Eq. 4), temporal error decreases monotonically — longer propagation chains are *smoother* — while accuracy degrades mildly and non-monotonically (0.4614 at $K{=}1$ to 0.4297 at $K{=}20$, with a recovery at $K{=}15$ that traces back to sequence-level variance rather than to a resonance effect). The knee of the consistency–latency trade-off sits at $K{\approx}10$, which we adopt as the default for adaptive scheduling; $K{=}3$ is the accuracy-preserving alternative.

4.5 Ablations: the improvement is structural

Table 5 isolates each design decision on Sintel; four conclusions hold.

(i) Temporal EMA filtering hurts — scene-dependently. Replacing pure re-anchoring with an exponential moving average ($\lambda{=}0.6$) between propagated and fresh fields re-injects 40% stale content at every keyframe. On the full 23-sequence protocol the aggregate damage is mild (kf10-adp: 0.4429 vs.

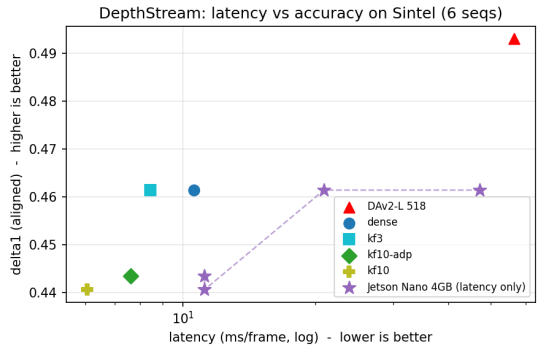


Figure 3: **Latency–accuracy plane** on Sintel. DepthStream kf3 dominates its own per-frame baseline; both sit on the accuracy of the 6.7× slower teacher with a fraction of the latency. Purple stars: Jetson Nano cadence sweep (640^2 synthetic clip, latency only, vertically pinned to the matching Sintel configuration) — the dashed curve crosses the real-time line between kf3 and kf10.

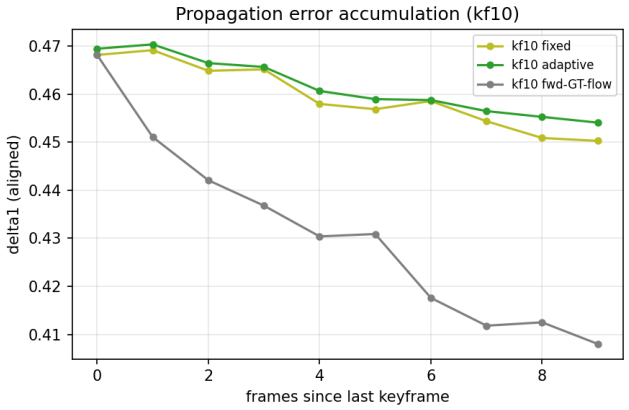


Figure 4: **Propagation decay.** δ_1 vs. frames since the last keyframe (Sintel). Adaptive scheduling flattens the decay to -3.3% over nine frames.

0.4434), but the six-sequence subset containing the dense-vegetation ambush sequences shows a large drop (0.2875 to 0.2772 at K=3): wherever occlusion is frequent, the stale propagated content is wrong, and blending spreads the error. Re-anchoring (Eq. 3) achieves continuity through the scale–shift alignment alone and is robust by construction, so we adopt it throughout.

(ii) Flow quality is not the bottleneck. Propagating with ground-truth *forward* flow scores *worse* than estimated backward DIS flow (0.4145 vs. 0.4406): the forward-flow coordinate-shift sampling approximation is biased at occlusions, whereas a backward flow is defined exactly at the output pixels. Doubling the flow resolution is likewise neutral (0.4619 vs. 0.4614). Propagation error is dominated by occlusion structure, not by flow accuracy — visible in the error maps of Fig. 6, where residual error concentrates at motion boundaries.

(iii) Adaptive scheduling buys back long-interval accuracy. At K=10, adaptive re-anchoring recovers +0.021 δ_1 over fixed scheduling (0.4434 vs. 0.4406) and attains the best temporal error of all configs (0.1054), at +1 ms. Notably, the photometric monitor is a *dormant safety net* on our benchmarks: it fires on only 1 of 1,064 Sintel frames and 0 of 540 Bonn frames at θ=0.15, η=0.30 — DIS flow explains almost all of the observed motion — yet a single re-timed keyframe rephases the cadence of its whole sequence and recovers +0.003 δ_1 on average. The monitor's insurance value should grow on production streams with cuts, motion blur, and exposure changes, which our laboratory benchmarks underrepresent.

(iv) Geometric (homography) propagation is a trap. A RANSAC homography decomposition — warp by the dominant static motion, flow only on the residual — improves the single rotation-dominated sequence (temple_2, +13% relative) but loses overall (0.2776 vs. 0.2827 mean δ_1 on a four-sequence pilot): Sintel's motion is dominated by forward translation, which a single homography cannot express, while dense backward flow covers it implicitly. We do not pursue

Table 2: **Per-sequence** δ_1 on all 23 Sintel sequences (GT-capped protocol, TRT fp16 backend).

Sequence	dense	kf3	kf10	kf10-adp
alley_1	0.5992	0.5985	0.5673	0.5673
alley_2	0.1741	0.1775	0.1796	0.1796
ambush_2	0.0263	0.0294	0.0327	0.0327
ambush_4	0.3564	0.3602	0.3033	0.3666
ambush_5	0.0760	0.0737	0.0766	0.0766
ambush_6	0.4611	0.4352	0.3449	0.3449
ambush_7	0.9920	0.9912	0.9908	0.9908
bamboo_1	0.5883	0.5877	0.5762	0.5762
bamboo_2	0.2989	0.3046	0.2989	0.2989
bandage_1	0.6911	0.6971	0.6823	0.6823
bandage_2	0.5193	0.5273	0.5196	0.5196
cave_2	0.1406	0.1408	0.1450	0.1450
cave_4	0.2565	0.2630	0.2310	0.2310
market_2	0.2462	0.2481	0.2350	0.2350
market_5	0.3585	0.3592	0.3264	0.3264
market_6	0.1140	0.1115	0.1128	0.1128
mountain_1	0.6276	0.6276	0.5998	0.5998
shaman_2	0.6095	0.6093	0.5897	0.5897
shaman_3	0.9737	0.9739	0.9700	0.9700
sleeping_1	0.9363	0.9376	0.9370	0.9370
sleeping_2	0.7923	0.8011	0.7706	0.7706
temple_2	0.1721	0.1701	0.1490	0.1490
temple_3	0.6030	0.5875	0.4958	0.4958

Table 3: **Per-sequence** δ_1 on the 9 Bonn RGB-D sequences (Kinect GT, TRT fp32 backend). crowd saturates the metric because its consecutive Kinect frames are near-duplicates.

Sequence	dense	kf3	kf10-adp
balloon	0.9999	1.0000	1.0000
balloon2	0.9368	0.9378	0.9105
balloon_tracking	0.9954	0.9958	0.9958
balloon_tracking2	0.9916	0.9918	0.9926
crowd	1.0000	1.0000	1.0000
crowd2	0.9686	0.9623	0.9434
crowd3	0.9999	0.9999	0.9999
kidnapping_box	0.9964	0.9967	0.9960
kidnapping_box2	0.9911	0.9910	0.9914

Table 4: τ_5 **temporal consistency**. Top: Sintel (23 seq; GT-flow warping). Bottom: Bonn (9 seq; estimated backward flow, identical across configs). Per-pair scale–shift alignment, 5% inlier threshold; higher is better.

Config	dense	kf3	kf10-adp
Sintel ($\tau_5\uparrow$)	0.5516	0.7332	**0.8303**
Bonn ($\tau_5\uparrow$)	0.8737	0.8786	**0.8854**

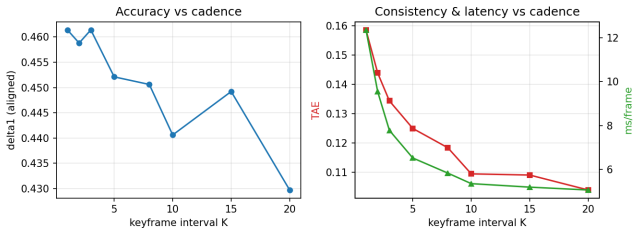


Figure 5: **Cadence sweep** (Sintel, 23 seq, TRT fp16). Left: accuracy degrades mildly and non-monotonically with K. Right: temporal error (TAE, GT flow) decreases monotonically while latency asymptotes to the flow cost — the knee sits at $K{\approx}10$.

this direction further.

4.6 Platforms and backend invariance

Table 6 extends the evaluation across devices, from a 2019 $99 developer board to a laptop GPU. The CPU runs the fp32 ONNX graph with 4 threads. Three claims matter for deployment: (i) *accuracy is backend-invariant* — over all configurations the fp32-CPU and fp16-GPU predictions agree to $\delta_1 \leq 10^{-4}$; (ii) *amortization flips the CPU calculus* — with the student's cost amortized over ten frames, the CPU stream is not merely real-time (249 FPS end-to-end at $640{\times}480$) but *faster* than the GPU stream, because flow and warping dominate the per-frame budget once inference is amortized away; and (iii) *2019 hardware streams in real time* — on a Jetson Nano 4GB (Maxwell GPU, 4 GB shared memory), per-frame inference reaches only 21 FPS at $640{\times}480$, below the real-time line, while DepthStream kf10 runs at 90 FPS ($4.2\times$; the full cadence sweep 21/48/90/90 FPS traces the amortization curve on Fig. 3) — the same runtime whose closest learned-propagation competitor [3] targets an RTX 3090. Cross-device agreement of the streamed fields (Nano TRT fp32 vs. desktop ORT fp32, 30 frames) is Pearson $r{=}0.956$; the residual gap reflects platform-specific engine kernels and flow-level numerics, both invisible to the affine-invariant protocol of Tables 1 and 5.

4.7 Analysis: when does selective refinement pay off?

A natural competitor to amortization is *selection*: run the large teacher only where the small model is expected to err. We built this speculative-cascade runtime (flip-consistency and edge cues select the top $p\%$ of $16{\times}16$ blocks; the teacher runs on the selected crops; crops are affine-aligned on their rings and feather-blended). On Sintel, the student already agrees with the teacher on 93.6% of pixels (δ_1 against the teacher as reference), and every cascade budget we tested (10/20/40%) *lost* accuracy against the pure student while costing $70\times$ more latency — crop-level ViT-L inference cannot amortize below a single full-image forward pass on this hardware. The negative result is informative: distilled students leave no disagreement space for a teacher to arbitrate, which is precisely why propagation-based systems over *non-distilled* foundation models [3] require learned corrections. Selection, not amortization, is the right tool for foundation-model students; amortization is the right tool for distilled ones.

Table 5: **Design-space analysis** (Sintel, 23 seq). "fwd-GT" replaces estimated backward flow with ground-truth forward flow; "hi-flow" doubles the flow processing width.

Variant	$\delta_1\uparrow$	TAE$\downarrow$	ms$\downarrow$
dense (per-frame)	0.4614	0.1585	10.6
kf3 (default)	0.4614	0.1343	8.4
kf3 + EMA (λ=0.6)	0.4614	0.1345	8.6
kf10 (fixed)	0.4406	0.1094	6.1
kf10-adp (adaptive)	0.4434	0.1054	7.6
kf10 + fwd-GT flow	0.4145	0.1282	6.3
kf3 + hi-flow (384)	0.4619	0.1354	9.4

EMA damage is scene-dependent: on the ambush (dense-vegetation) subset it drops δ_1 from 0.2875 to 0.2772 (K=3).

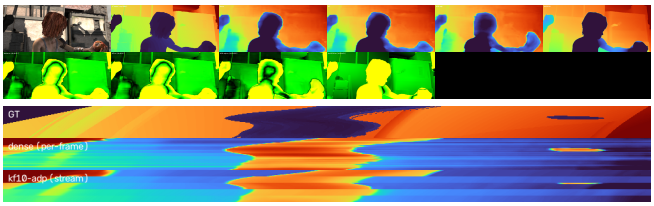


Figure 6: **Qualitative comparison** (Sintel alley_1). Top: RGB, ground truth, and affine-aligned predictions of dense, kf3, kf10-adp, and the teacher, with error maps (δ_1 inset; green = accurate, red = $>25\%$ relative error). Bottom: **space-time slice** — depth along one scanline stacked over 50 frames. Per-frame inference (middle) flickers; the stream (bottom) forms smooth bands.

4.8 Negative result: homography-based geometric propagation

We additionally implemented a geometric variant: RANSAC homography on sparse flow correspondences, homography warping on static-inlier regions, flow elsewhere. It improves the single rotation-dominated sequence (temple_2, $+13\%$ relative δ_1) but loses overall (0.2776 vs. 0.2827 mean on a four-sequence pilot): Sintel's camera motion is dominated by forward translation with strong parallax, which a single global homography cannot represent, while dense backward flow

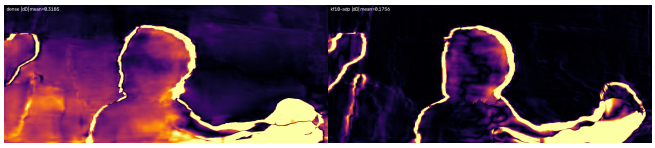


Figure 7: **Frame-to-frame flicker** $|D_t - \mathcal{W}(D_{t-1}, F^{gt})|$ (Sintel, GT-flow warp). Streaming reduces the mean flicker by 45% (0.3185 → 0.1756); residual error concentrates at motion boundaries — the expected failure mode of flow propagation.

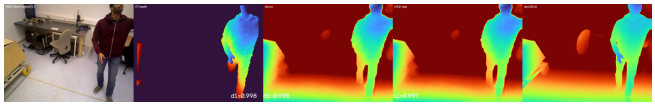


Figure 8: **Real data** (Bonn balloon2): RGB, Kinect ground truth, and affine-aligned predictions. δ_1 insets: 0.998 / 0.998 / 0.997.

covers translation and rotation uniformly. Dense learned-free flow is a strong propagator precisely because it makes no motion-model assumption.

5 Discussion

5.1 DepthStream vs. learned propagation

Table 8 summarizes the contrast with the closest system. The two approaches are complementary rather than competing: VeloDepth demonstrates that learned corrections squeeze extra consistency out of a *non-distilled* foundation base, while DepthStream shows that for students distilled from the judge model, the same skeleton — keyframe, warp, re-anchor — delivers a large share of the benefit with no training pipeline, an order of magnitude lower latency, and 7× smaller hardware. The practical decision rule that emerges from our analysis (Sec. 4.7) is: *if your tiny model is distilled from your quality reference, propagate — do not correct; and re-anchor — do not smooth.*

5.2 When does amortization stop paying?

Three boundaries emerged from our experiments. *Occlusion density*: propagation error concentrates at disocclusions (Fig. 7); adaptive scheduling hides the symptom at scene level but a per-pixel mask with fallback inference is the natural next step. *Extreme intervals*: beyond $K{\approx}15$ the accumulation curve (Fig. 4) falls below the dense baseline, so the cadence should stay within the range where decay is flatter than the model's own frame-to-frame variance — empirically $K{\leq}10$ here. *Model mismatch*: if the runtime's student is *not* distilled from the referee model, a disagreement space re-opens and selective correction (Sec. 4.7) or learned propagation [3] becomes rational again.

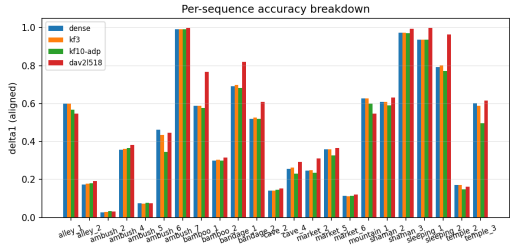


Figure 9: **Per-sequence accuracy** on Sintel (six-sequence protocol). Sequence-level variance is dominated by scene difficulty (dense vegetation in ambush sequences), not by the streaming mechanism.

Table 6: **Platforms** (student backend; E2E on a 640×480 synthetic clip, kf10, MAXN power mode). Nano cadence sweep (per-frame/kf3/kf10/kf10-adp): 47.2/20.9/11.2/11.2 ms = 21.2/47.9/89.6/89.6 FPS (see also Fig. 3). Backend invariance: $|\Delta\delta_1| \leq 10^{-4}$ between GPU-fp16 and CPU-fp32 on the Sintel protocol.

Platform	Backend	Prec.	model	E2E kf10
RTX 3060 Laptop 6GB	TensorRT 11.3	fp16	1.20 ms	∼150 FPS
Ryzen 7 5800H (CPU)	ORT 1.26	fp32	6.95 ms	249 FPS
Jetson Nano 4GB (2019)	TensorRT 8.2	fp32	29.9 ms	90 FPS

6 Limitations

(i) Flow propagation fails at disocclusions and large rotations; the adaptive scheduler masks the symptom (early re-anchoring) but does not model occlusion explicitly. (ii) Our evaluation covers one synthetic and nine real sequences; extending Bonn to all published sequences and adding TUM would strengthen the real-data claim. (iii) Power consumption is not yet reported for the platform table, and GPU-side evaluation uses one consumer laptop GPU; Orin-class Jetson results are expected to transfer directly (same TensorRT pipeline, official DepthART scripts exist). (iv) The student consumes no camera intrinsics here; the metric variant of DepthART would require propagating K jointly with the field. (v) VeloDepth [3] reports a temporal-consistency metric (τ_5^{SSI}) that we have not yet reproduced, so our comparison to it is architectural and latency-based rather than metric-exact.

7 Conclusion

Streaming depth, formulated as anchored-field maintenance, changes what a tiny model can do: at $K{=}3$ the stream matches per-frame accuracy exactly while lowering latency and raising temporally consistent pixels from 55% to 73%, on synthetic and real benchmarks alike. The error decomposition explains both the gain and its limits — the cadence knee at $K^{*}{\approx}10$, the occlusion bound on transport error, and the distillation closure that leaves no room for learned correction on distilled students. Operationally, the formulation moves real-time video depth

Table 7: **Per-component budget** (ms; 640×480). model = one student call as executed by the runtime (preprocess + host–device transfer + engine); flow = one DIS field at 256 px incl. upsampling; warp = full-resolution remap. At 224^2 the Python-side preprocessing dominates the model call on every platform.

Platform	model	flow	warp
RTX 3060 Laptop (TRT fp16)	12.4	2.2	0.8
Ryzen 7 5800H (ORT fp32)	12.1	2.2	0.7
Jetson Nano (TRT fp32)	46.6	7.3	1.0

Table 8: **DepthStream vs. learned propagation** (VeloDepth numbers from their paper; both on 0.3–0.5 MP inputs).

	VeloDepth	DepthStream (ours)
propagation module	55 M, trained	none
training cost	8×4090, 6 days	—
base model	409 M (UniDepthV2)	6 M distilled
keyframe trigger	learned gating	residual (free)
hardware	RTX 3090	RTX 3060 Laptop 6GB
throughput	26.2 FPS	142–176 FPS

onto hardware where the per-frame formulation cannot go: 90 FPS on a 2019 $99 board, 249 FPS on a CPU, identical accuracy on either backend. The practical rule follows from the decomposition: propagate — do not correct; re-anchor — do not smooth.

Reproducibility and licensing

All tables are produced by single commands (`eval_sintel_v2.py`, `eval_bonn.py`, `bench_cpu.py`; figures by `plots.py`, `qual_figs.py`, and `draw_framework.py`). The student checkpoint and teacher weights are public; the student and parts of its supervision inherit CC BY-NC 4.0 terms from DepthART and Depth Anything V2, which restricts commercial use of the released weights (the runtime itself has no such restriction). Sintel and Bonn are used under their respective research licenses.

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